ENG101
LESSON 21

Flowcharting:

So far we have dealt mainly with computers, but now it is imperative that we find out how a program is written. In all activities involving computers, it is necessary that the programmer is aware of what the machine is doing and what a program is supposed to do. As previously mentioned flowcharting, one of the steps in programming, indicates the logical path the computer will follow in executing a program; it is a drawing very much like a road map. Flowcharting is not restricted to the preparation of program sin a particular language and should be done for each major problem before the writing of the program is attempted.

If the finished program does not run as it should, the errors are more easily detected on the flowchart than in the maze of words, characters, and numbers that make up the computer program. In order to develop a flowchart successfully, a programmer should be aware of the sequence of steps needed to obtain a correct solution to a problem.

There are two ways of making a flowchart: the freehand version and the neater, more readable version. In the former version, the graphic outlines are simply jotted down as the steps of the program are worked out. This is quite satisfactory if the flowchart is not intended to be kept as a permanent record. However, if a permanent, neater and more readable flowchart is needed, the latter

Method whereby a template, a sheet of plastic with all the flowcharting symbols cut into it, is used.

The following symbols should be used for the purpose of uniformity. The first and last sy is . This is terminal symbol which indicates the beginning or the end of a program. The word 'START' must be inserted inside the figure if it is the beginning of the program and 'STOP' if it is the end of the program. The figure in the form of a parallelogram is used as an input/output symbol. It indicates that something is either brought to or taken form the am. The rectangular symbol stands for processing and

Indicates a place in the program where action is taken. In a program, to indicate that a decision has to be made, the diamond shaped symbol is used. The decision is usually in the form of a question that must be answered by either 'yes' or 'no'. Finally, the arrows are used to show that the flow or direction in which the different actions in the program are performed.

It should be noted that flowchart is not a program, but only a step in the preparation of a program, and is used in determining how to step up and write the program. However, if the problem is not understood,

neither the flowchart nor the program can be done correctly. It is possible for two programmers working separately, to write programs to solve the same problem and come up with flowcharts and programs that may be altogether different.

After a program has been worked out, it is usually written down and kept with a copy of the flowchart along with detailed instructions for the used and interpretation of the program. This procedure is part of what is referred to as program documentation. If documentation isn't available, it is always possible to work backwards and make a flowchart from an application program.
It may be necessary to create a new flowchart when the original one is missing, in order to understand the program for which it was a preparatory step.
Flowcharting is one of the first things a student programmer is taught, because a flowchart shows how a person thinks about a problem. In other words it is through this that a new programmer reveals his or her logical and analytical ability, which is a must in programming.

Exercises

1. Main Idea

Which statement expresses the main idea of the text? Why did you eliminate the other choices?

1. Every programmer must know how to flowchart.
2. Program documentation specifies what the program is supposed to do.
3. Flowcharting is a basic step in programming.

2. Understanding the Passage

Decide whether the following statements are true or false (T/F) by referring to the information in the text. Then make the necessary changes so that the false statements become true.

☐ 1. A good flowchart takes into account the steps which are necessary to solve the problem.

☐ 2. It is not possible to draw a flowchart without using a template.

☐ 3.There is only one possible flowchart for every problem.

☐ 4.Every programmer must learn flowcharting and realize its importance.

☐ 5.The method of flowcharting depends on the programming language being used.

☐ 6.Flowcharts show the logic one has to follow to solve problem.

☐ 7.Documenting a program is essential in explaining what the program is supposed to do.

☐ 8.If the flowchart is correct, the program will certainly work.

☐ 9.Each symbol in flowcharting has a specific meaning.

☐ 10.Flowchart can show processes, but not decision.

3. Locating Information

………..1. A programmer must document his program in order that others may be able to understand it.
………..2. Flowcharting resembles a map.
………..3. Flowcharting shows the logical ability of a programmer.
………..4. There is more than one way of flowcharting.
………..5. A certain symbol is used to indicate if a question is to be answered 'yes' or 'no'.

4. Contextual Reference

Look back at the text and find out what the words in bold typeface refer to.
1.does not run as it should (l.10)……………….
2.In the former version (l.16)……………….

3.This is quite satisfactory (l.18)……………….
4.the latter method (l.20)……………….
5.flowcharting symbols cut into it (l.21)………………
6.which indicates the beginning (l.23)………………
7.It indicates that something (l.27)……………….
8.that may be altogether (l.40)……………….different
9.the original one is missing (l.48)………………
10.which is a must in programming (l.53)………………

5. Understanding Words

Refer back to the text and find synonyms for the following words.

1. route (l.6) ……………………
2. try (1.9)……………………
3. answer (l.14)…………………..
4. put in (l.25)…………………..
5. show (l.52)…………………..

Now refer back to the text and find antonyms for the following words.

6. unlimited (l.7) ……………………
7. undiscovered (1.11)……………………
8. temporary (l.19)…………………..
9. inaccessible (l.45)…………………..
10. illogical (l.53)…………………..

6. Word forms

First choose the appropriate form of the words to complete the sentence. Then check the differences of meaning in your dictionary.

1.involvement, involve, involved, involving

a. In most operations ……………………. calculations, computers can do the job much faster than man.

b. Flowcharting …………………… a logical analysis of a problem and a diagrammatic representation of the sequence of events to be followed in solving the problem.

c. The …………………… of the new programmer in the user’s group was appreciated by his manager.

2. correction, correct, corrected, corrective, correcting
a.It is always a good approach to …………………. The errors in your program before running it with the data.
b.In order to develop a good flowchart, a programmer should be aware of the sequence of steps needed to obtain a ……………………… solution to a problem.
c.He submitted the ……………………….. version of the program to be keypunched.
3.Process, processed, processing
a.The Central ……………………….. Unit is responsible for executing the programs.
b.A block diagram can show if a ………………….. Has to be repeated or if there are alternative routes to be taken.
c.The applications of all the new students were ………………………. by the computer.
4. performance, perform, performed, performing, performer
a.……………………….. is a verb used quite often in COBOL programming.
b.Data processing refers to the operations which are …………………….. on the data either to derive information from them or to order them in files.
c.The …………………… of the computer salesman was measured by the number of units he sold.
5.documentation, document, documented, documenting
a.……………………… a program is essential so that other programmers can understand it.
b. It took the programmer one week to complete the …………………… of the programs in the new system.
c. The payroll package we purchased is very well…………………………….

7a Content Review

Try to think of a definition for each of these items before checking them in the Glossary. Then complete the following statements with the appropriate words. Make sure you use the correct form, i.e. singular or plural.

executing terminal rectangle
template parallelogram diamond
documentation

1. The information describing what a program can do and what a program can do and what the results mean is referred to as ……………..……………..
2. It is advisable to test the program without data before………………………………..…it.
3. A piece of plastic with different shapes used for flowcharting is called a ……………….
4. Data used as input must be indicated with a ………………………………….
5. The symbol which marks the beginning and end of a program is called the …….. symbol.

Example

1. At the rate computer technology is growing, today's computers might be obsolete by 2005and most certainly by 2010.

CONDITION	PREDICTION
At the rate computer technology is growing	today's computers might be obsolete by 2005 and most certainly by 2010

2. If we couldn't feed information in and get results back, computers wouldn't be of much use:

CONDITION	PREDICTION
If we couldn't feed information in and get results back,	computers wouldn't be of much use

3. If the hammer in drum printers hits a little early or late, the characters will appear slightly above or below the line.

CONDITION	PREDICTION
If the hammer in drum Printers hits a little early or late,	the characters will appear slightly above or below the line

Exercise

Read the following sentences and underline the part that expresses a condition, once; and the part expressing a prediction, twice.

1. It has been said that if transport technology had developed as rapidly as computer technology, a trip across the Atlantic Ocean today would take a few seconds.
2. Working for the U.S. Census Bureau, Dr. Hollerith realized that unless some means of speeding up the analyses of census data were found, it would take more than ten years to complete the job.
3. If the hammer in train printers hits a little early or late, the character will appear slightly to the right of its proper position.

4. Mainframes would still be occupying a lot of space if it weren’t for microminiaturization.
5. If computer technology continues growing at the rate it has, bubble memory will soon replace the chip.